# SWA-DPBR: Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability for Cross-Cultural Moral Alignment of Large Language Models

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## Abstract

Deployed LLMs still moralise like a single culture even when users are not. Fine-tuning and culture-specific labels do not scale to hundreds of countries, so we stay at inference time and ask what public survey data can buy. **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability) keeps the WVS-grounded persona panel and Prospect-Theory-weighted importance sampling over logit perturbations from our earlier SWA pipeline, but replaces a single batch of $K{=}128$ samples with *two* independent halves of size $K/2$. The squared gap between the two IS estimates acts as a bootstrap-style variance proxy; a soft reliability weight $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ down-weights the correction when the two passes disagree, and an ESS-adaptive blend pulls unstable estimates toward the debiased base logit before the hierarchical country prior runs. We report a fixed 20-country MultiTP slice (ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM) and macro-averaged misalignment score (MIS, $\ell_2$ distance between human and model AMCE vectors; lower is better) for eight open models from 270M to 70B. Phi-4 reaches mean MIS $0.35$ with an $18/20$ country win rate versus vanilla ($+23.7\%$ relative reduction in macro MIS); Qwen3-VL-8B and Llama-3.3-70B win every country in this panel ($20/20$) with $+18.9\%$ and $+21.4\%$ macro gains. Tiny Gemma-3-270M barely moves the aggregate ($+0.3\%$), while Gemma-7B in 4-bit drifts slightly negative—evidence that quantisation and capacity interact sharply with IS noise. Code: [repository].

## 1 Introduction

When a self-driving car must choose between sparing a group of elderly men or a group of young women, there is no culturally universal answer. The Moral Machine experiment [Awad et al., 2018] collected 40 million judgments from 233 countries and revealed that such preferences vary profoundly across cultures: East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to assist with such decisions encode a single, culturally monolithic moral stance inherited from their predominantly Western training data [Henrich et al., 2010].

The multilingual successor benchmark MultiTP [Jin et al., 2025] operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages, providing a rigorous test bed for cross-cultural alignment. The verdict is stark: Jin et al. [2025] found that none of 19 contemporary LLMs achieved broadly satisfactory alignment; most exhibit WEIRD-biased reasoning that departs markedly from non-WEIRD human preferences.

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How should this gap be closed? Existing approaches each sacrifice critical properties. *Fine-tuning* [Bai et al., 2022, Ouyang et al., 2022] demands per-culture labelled data and modifies weights irreversibly. *Prompting-based steering* relies on ad-hoc instructions with no empirical grounding [Jin et al., 2025]. *Activation steering* [Arditi et al., 2025, Zou et al., 2023] requires white-box access and per-model engineering. *Reward-guided decoding* [Khanov et al., 2024, Mudgal et al., 2024] needs trained reward models unavailable for cross-cultural settings. *Culture-aware adapters* [Li et al., 2024, Yao et al., 2024] scale poorly across 100+ countries. None of these methods can adapt an LLM's moral reasoning to a target culture *at inference time* without labelled data or weight modification.

We propose **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability). The organising idea is unchanged: *disagreement is signal*. Moral preferences within any society are not monolithic—age cohorts and social strata exhibit systematically different priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] provides empirically calibrated data on these within-country differences across 80+ nations. When culturally grounded persona agents derived from WVS data *disagree* on a dilemma, that disagreement marks where a controlled logit nudge may help rather than defaulting to a single answer.

Concretely, SWA-DPBR constructs $N = 4$ WVS-grounded personas per country (three age cohorts plus a utilitarian agent), evaluates all $N + 1$ agents in one batched forward pass with native-language prompting, and extracts decision logits. It then runs *two* independent Prospect-Theory-weighted importance-sampling passes, each with $K/2{=}64$ perturbations (total Monte Carlo budget $K{=}128$ unchanged). Writing $\delta_1^\star, \delta_2^\star$ for the debiased IS corrections from the two passes, we set bootstrap variance $v_\mathrm{b} = (\delta_1^\star - \delta_2^\star)^2$ and soft reliability $r = \exp(-v_\mathrm{b}/s)$ with scale $s$ (hyperparameter `VAR_SCALE`). The blended correction is $r \cdot (\delta_1^\star + \delta_2^\star)/2$, so unstable pairs of IS estimates are automatically attenuated. When effective sample size in either pass is low, an ESS-adaptive anchor regulariser interpolates the micro-correction toward the debiased base logit before the hierarchical country prior updates [Kahneman and Tversky, 1979]. Positional debiasing cancels recency bias as in the original pipeline.

**Contributions.**

- **A training-free framework** (§3) that turns within-country WVS variation into a steering signal: a Prospect-Theory-weighted dual-pass IS correction with bootstrap reliability damping on every scenario (EXP-24 implementation in `experiment_DM/exp24_dpbr_core.py`).
- **Fresh 20-country, eight-model MIS sweep** (§5) spanning 270M to 70B parameters, including a vision-language backbone, with per-country win rates against vanilla and macro MIS summaries.
- **A mechanism analysis backed by a robustness suite** (§6, Appendices E–P). Table 6 isolates IS, debiasing, and the persona panel; deterministic consensus-shift baselines and Binary-ARGS (Appendix N) show that WVS-only linear rewards underperform the Prospect-Theory kernel. Three temperature sweeps and six preprocessing configs on the archival 15-country panel stay inside the bootstrap noise floor.

## 2 Related Work

**Cross-cultural moral benchmarks.** The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs; we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-DPBR targets binary moral decisions specifically.

**Cross-cultural LLM evaluation.** LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-DPBR's calibration objective.

**Inference-time alignment.** Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models unavailable for cross-cultural settings (Appendix H). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. To the best of our knowledge, SWA-DPBR is the first training-free method targeting country-level moral calibration specifically. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-DPBR operationalises concretely.

**Positioning against adjacent inference-time methods.** Recent work on inference-time alignment falls into several families that we deliberately do not compete with head-to-head. *DIFFPO*-style policy-agnostic aligners require an additional learned aligner model on top of the base LLM, trading deployment cost for flexibility; *SWAI*-style training-free logit steering targets attribute-level shifts (style, toxicity) via fixed steering vectors, not country-conditioned moral preferences; *SLED*-style self-logit evolution is a white-box factuality method that operates on internal layers. SWA-DPBR is positioned narrowly within this design space: black-box, single-model, binary forced-choice moral decisions with country-conditioned persona grounding, and no extra trained aligner. Relative to structured multi-agent pipelines such as OG-MAR [Zhang et al., 2024], SWA-DPBR trades ontology-level reasoning depth for the much lower deployment cost of one additional batched forward pass.

**Importance sampling, reliability, and Prospect Theory.** Self-normalised importance sampling is biased at finite $K$; splitting the same Monte Carlo budget into two independent replicates and comparing their outputs is a standard way to expose sampling noise without doubling wall-clock forward passes at the transformer level (each replicate reuses cached persona logits). We combine that diagnostic with a Prospect-Theory utility [Kahneman and Tversky, 1979] on logit-gap coordinates, following the control-as-inference view used in path-integral and KL-regularised steering work [Williams et al., 2018, Huang et al., 2024]. Dual-pass reliability is a variance-based heuristic, not an unbiasedness certificate; we report it as part of the method's explicit uncertainty model.

## 3 Method: SWA-DPBR

### 3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` ($a$) and `B` ($b$). For target country $c$, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM $f_\theta$ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. For the five binary dimensions (Species, Gender, Age, Fitness, Social Value), this reduces to the empirical mean of $p_{\text{spare}}$. For Utilitarianism the predictor is the count difference $n_{\text{diff}} = n_{\text{pref}} - n_{\text{non-pref}}$; we fit $p_{\text{spare}}(\mathbf{x}) = a + b\, n_{\text{diff}}(\mathbf{x}) + \varepsilon$ by OLS and report $\hat{m}_{\text{Util}}^{(c)} = a + b\, \overline{n_{\text{diff}}}$, i.e. the fitted value at the mean observed count difference (Appendix A).

**Goal:** Without modifying $\theta$, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$ under Jensen–Shannon Divergence.

### 3.2 Cultural Persona Construction

For each country $c$, SWA-DPBR constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. We extract 8 value dimensions from specific WVS items: *gender egalitarianism* (Q58P, Q59P, Q60P), *religious importance* (Q6P), *interpersonal trust* (Q43P), *moral permissiveness* (Q50, Q52P, Q54P), *work centrality* (Q5P), *family importance* (Q1P), *personal autonomy* (Q39P), and *meritocratic orientation* (Q40P). For each country, respondents are segmented

into three age cohorts using birth year and survey year: *young* (18–35), *middle-aged* (36–55), and *older* (>55). Dimension means are computed per cohort and converted into natural-language descriptors via fixed thresholds (Appendix B). A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a natural-language system prompt:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

### 3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures $T_{\text{cat}}$ (Table 1) scale logits before computing decision gaps:

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here $\delta_{\text{base}}$ is the decision gap of the base prompt (no persona). The reward $r_i$ measures how strongly persona $i$ shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in EXP-24 (no $\tau$ threshold in `experiment_DM/exp24_dpbr_core.py`).

### 3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A ↔ B) and group labels (Group A ↔ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields $\delta_{\text{base}}$ under the same rule.

### 3.5 Prospect-Theory IS with Dual-Pass Reliability

SWA-DPBR runs two *independent* importance-sampling passes on every scenario. Each pass draws $K_{\text{half}} = K/2 = 64$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; $\sigma$ defaults to $0.3$ but is lifted to at least the empirical standard deviation of persona gaps when that is larger (implementation matches `exp24_dpbr_core.py`). Candidates are $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

**Per-persona gains and collective utility.** For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, define *scalar gains* in logit-gap units that measure how much moving the consensus toward $\tilde{\delta}_k$ reduces $L^1$ distance to each persona relative to the base model:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{5}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{6}$$

Large $g_{i,k}$ means $\tilde{\delta}_k$ is closer than $\delta_{\text{base}}$ to persona $i$'s debiased gap; $g_{\text{cons},k}$ is the same idea with the persona consensus $\bar{\delta}$ as the reference. Prospect Theory is applied *per persona before averaging*, preserving loss aversion at the social level [Kahneman and Tversky, 1979]:

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}})\,\frac{1}{N}\sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big)\ +\ \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big)\,, \tag{7}$$

Table 1: SWA-DPBR hyperparameters. PT parameters from Kahneman and Tversky [1979]; others validated on a held-out synthetic set with author proxy labels (Appendix D).

| Parameter | Symbol | Value | Source |
|---|---|---|---|
| Persona agents | $N$ | 4 | Design |
| IS samples / pass | $K_{\text{half}}$ | 64 | Design (2× for $K$=128) |
| Reliability scale | $s$ (`VAR_SCALE`) | 0.04 | EXP-24 default |
| Perturbation $\sigma$ | $\sigma$ | 0.3 | Validation |
| Cooperation weight | $\lambda_{\text{coop}}$ | 0.7 | Validation |
| IS softmax temp. | $\eta$ | 0.5 | Validation (`temperature`) |
| PT curvature | $\alpha$=$\beta$ | 0.88 | Kahneman and Tversky [1979] |
| PT loss aversion | $\kappa$ | 2.25 | Kahneman and Tversky [1979] |
| Decision temp. | $T_{\text{dec}}$ | 0.5 | Validation |
| $T_{\text{cat}}$ (Species / Gender / Others) | | 4.0 / 3.5 / 1.5 | Category |

with $\lambda_{\text{coop}} = 0.7$. Dividing by $\sigma$ makes the PT input dimensionless and comparable across moral dimensions (matches `src/controller.py` and `experiment_DM/exp24_dpbr_core.py`). The per-agent offsets $r_i = \delta_i - \delta_{\text{base}}$ from Eq. 4 remain useful diagnostics but **do not** enter $U_{\text{total}}$ in this implementation.

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\,|x|^{\beta} & x < 0. \end{cases} \tag{8}$$

We use $\alpha = \beta = 0.88$ and $\kappa = 2.25$ [Kahneman and Tversky, 1979]. There is **no** separate quadratic penalty on $\epsilon_k$ in the released utility: the Gaussian proposal $\mathcal{N}(0, \sigma^2)$ already limits step size, and an explicit KL-on-$\epsilon$ term is omitted (see Appendix C).

**Per-pass IS update and ESS guard.** For each pass $m \in \{1, 2\}$ we form softmax weights over its own $K_{\text{half}}$ samples,

$$w_k^{(m)} = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta_m^{\star} = \sum_{k=1}^{K_{\text{half}}} w_k^{(m)}\,\epsilon_k, \tag{9}$$

with softmax inverse temperature $\eta = 0.5$ (field `temperature` / `mppi_temperature` in `SWAConfig`; distinct from $T_{\text{dec}}$ scaling the final sigmoid, though we also set $T_{\text{dec}} = 0.5$). Let $\text{ESS}^{(m)} = K_{\text{half}} / \sum_k (w_k^{(m)})^2$ and $\widehat{\text{ess}}^{(m)} = \text{ESS}^{(m)}/K_{\text{half}}$. If $\widehat{\text{ess}}^{(m)} < \rho_{\text{eff}}$ we set $\delta_m^{\star} \leftarrow 0$ (binary guard; default $\rho_{\text{eff}}$ matches the codebase).

**Bootstrap reliability and anchor blend.** Define bootstrap variance $v_{\text{b}} = (\delta_1^{\star} - \delta_2^{\star})^2$ and soft reliability $r = \exp(-v_{\text{b}}/s)$ with scale $s$=0.04 (`VAR_SCALE`). The merged IS offset is

$$\delta_{\text{IS}}^{\star} \;=\; r \cdot \frac{\delta_1^{\star} + \delta_2^{\star}}{2}. \tag{10}$$

Large disagreement between passes shrinks $r$ and pulls the correction toward zero without a hard gate. Finally, with $\text{ess}_{\min} = \min(\widehat{\text{ess}}^{(1)}, \widehat{\text{ess}}^{(2)})$ and lower clip $\rho_{\text{eff}}$, set $\alpha_{\text{reg}} = \text{clip}(\text{ess}_{\min},\, \rho_{\text{eff}},\, 1)$ and form the micro correction before the hierarchical prior,

$$\delta_{\text{micro}} = \alpha_{\text{reg}}\,\bar{\delta} + (1 - \alpha_{\text{reg}})\,\delta_{\text{base}} + \delta_{\text{IS}}^{\star}, \tag{11}$$

where $\delta_{\text{base}}$ is the symmetric swap average for the base prompt (§3.4). Setting `EXP24_ESS_ANCHOR_REG=0` recovers the legacy $\bar{\delta} + \delta_{\text{IS}}^{\star}$ path. The country prior (EXP-09) then applies an EMA blend to $\delta_{\text{micro}}$ as before. The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; Table 1 lists all hyperparameters (validation details in Appendix D).

**Algorithm 1** SWA-DPBR Inference for Scenario $\mathbf{x}$ in Country $c$

**Require:** Frozen LLM $f_\theta$; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K_{\text{half}}, \sigma, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, s$
**Ensure:** $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ §3.4
Compute consensus δ̄, rewards r_i                                   ▷ Eqs. 3–4
for m ∈ {1, 2} do
    for k = 1 to K_half do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eqs. 5–7
    end for
    δ*_m ← Σ_k w_k^(m) ε_k or 0 if ess̃^(m) < ρ_eff                  ▷ Eq. 9
end for
r ← exp(−(δ*_1 − δ*_2)²/s); δ*_IS ← r(δ*_1 + δ*_2)/2                ▷ Eq. 10
δ_micro ← α_reg δ̄ + (1 − α_reg) δ_base + δ*_IS                      ▷ Eq. 11
δ_final ← hierarchical prior on δ_micro
return p_spare = σ(δ_final / T_dec)
```

**Figure placeholder:** SWA-DPBR Pipeline Overview

Left: WVS Persona Construction → Middle: Batched Forward Pass → Right: Dual-pass PT-IS + reliability merge

Figure 1: **SWA-DPBR pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** Two independent IS passes merged with bootstrap reliability $r$ (Eq. 10).

# 4 Experimental Setup

## 4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 2). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix F).

Table 2: The six moral dimensions in MultiTP.

| Dimension | Preferred | Contra | $T_{\text{cat}}$ |
|---|---|---|---|
| Species | Human | Animal | 4.0 |
| Gender | Female | Male | 3.5 |
| Age | Young | Elderly | 1.5 |
| Fitness | Fit | Unfit | 1.5 |
| Social Value | High (exec.) | Low (homeless) | 1.5 |
| Utilitarianism | More lives | Fewer lives | 1.5 |

### 4.2 Models

Eight open-weight checkpoints span capacity regimes: **Phi-4**, **Phi-3.5-mini**, **Llama-3.2-1B**, **Llama-3.3-70B**, **Gemma-3-270M-IT**, **Gemma-4-2B-IT** (E2B), **Gemma-7B-IT** (4-bit), and **Qwen3-VL-8B**. Runtimes follow the EXP-24 driver (`exp_model/exp_24/*.py`): Unsloth or native HF depending on the family, greedy decoding, seed 42, and always-on dual-pass IS ($2\times64$ samples per scenario).

**Native-language evaluation.** Each country's scenarios are presented *entirely in the native language*: scenario framing, character descriptions, lane labels, and closing questions are all translated into 12 languages (zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es, en). Only the *decision tokens* (`A/B`) remain English; these are verified as single-token entries in each model's vocabulary (Appendix G). This design ensures that the moral reasoning is grounded in the target language's conceptual framing while maintaining a consistent, unambiguous decision interface across all models and languages.

### 4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix H). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix H). Primary headline metric is **MIS** (misalignment score): the $\ell_2$ distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], matching `src/amce.py`. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** $r$ and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla$\rightarrow$DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean $r$. Legacy 15-country JSD-centric tables in Appendix P pre-date EXP-24 and are kept for continuity with earlier drafts.

## 5 Results

### 5.1 Cross-Model Comparison (EXP-24, 20 Countries)

Table 3 compresses the EXP-24 / SWA-DPBR sweep that mirrors `exp_paper/tracker.md`. No single scaling law emerges: Phi-4 sits lowest in absolute MIS ($0.35$ macro mean) and shaves nearly a quarter off vanilla, yet it still loses ground on Iran and Brazil where vanilla was already competitive. Llama-3.3-70B starts from a much worse vanilla profile (mean MIS $0.85$) and drops to $0.67$—a $+21\%$ relative gain with a clean $20/20$ win slate—without delivering the smallest absolute MIS in the panel. Qwen3-VL-8B matches that win rate with a mid-$0.46$ MIS. At the opposite extreme, Gemma-3-270M barely beats vanilla in aggregate ($+0.3\%$) despite winning $16/20$ countries: the macro mean is dominated by a handful of large per-country misses, a useful reminder that win counts and macro averages tell different stories when variance is heavy-tailed. Gemma-7B in 4-bit is the only row with a negative macro delta ($-0.3\%$); quantised logits and IS variance appear mis-matched there.

**Reference.** The strongest single-configuration baseline recorded in our internal EXP-09 log is mean MIS $\approx 0.398$; averaging the eight rows above yields $\approx 0.49$, dominated by the 70B and mini-model

Table 3: EXP-24 macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean $\ell_2$ AMCE distance (↓). "Win" counts countries where DPBR MIS is strictly lower than vanilla. Mean $r$ averages the per-country Pearson correlation between model and human AMCE vectors.

| Model | Mean MIS ↓ | vs. van. (%) | Win / 20 | Mean $r$ |
|---|---|---|---|---|
| Phi-4 | **0.346** | +23.7 | 18 | +0.56 |
| Qwen3-VL-8B | 0.466 | +18.9 | **20** | +0.22 |
| Gemma-4-2B-it (E2B) | 0.479 | +3.4 | 14 | +0.46 |
| Gemma-3-270M-it | 0.478 | +0.3 | 16 | +0.25 |
| Gemma-7B-it (4-bit) | 0.459 | −0.3 | 7 | −0.48 |
| Llama-3.2-1B | 0.486 | +1.2 | 13 | −0.25 |
| Phi-3.5-mini | 0.571 | +10.4 | 17 | −0.35 |
| Llama-3.3-70B | 0.668 | +21.4 | **20** | −0.01 |

Table 4: Archival baseline comparison on Llama-3.1-70B (15-country MultiTP subset, single-pass SWA-PTIS before EXP-24). Included to show prompting/steering gaps; the DPBR numbers in Table 3 use Llama-3.3-70B on 20 countries.

| Method | Mean JSD ↓ | Pearson $r$ | vs SWA (%) | $\lvert\Delta\rvert$ |
|---|---|---|---|---|
| **SWA-PTIS (archival)** | **.0612 ± .004** | **0.384** | – | – |
| Country-Tailored (B1) | .0680 ± .004 | 0.233 | +11.1% | .0068 |
| PRISM-Style (B3) | .0699 ± .005 | 0.148 | +12.5% | .0087 |
| WVS Profile (B2) | .0737 ± .005 | 0.155 | +16.9% | .0125 |
| Activation Steering | .0877 ± .006 | −0.208 | +30.2% | .0265 |
| Vanilla | .0937 ± .006 | −0.026 | +34.7% | .0325 |

tails. Treat that gap as evidence that scale and instruction-tuning interact with DPBR, not as a claim that DPBR beats every historical ablation on every metric.

### 5.2 Baseline Comparison on Llama-3.1-70B (Archival)

Single-pass SWA-PTIS remains the strongest entry on Llama-3.1-70B in the older 15-country JSD study (Table 4), beating the strongest prompting baseline by an absolute $0.0068$ JSD. Activation steering performs *below* vanilla, suggesting that contrastive steering vectors interfere with moral reasoning on that benchmark rather than helping it.

**Isolating the active mechanism (same panel).** Stripped-down baselines that share the batched multi-agent forward pass and positional debiasing but swap the decoding rule show that deterministic consensus shifts close most of the PT-IS gap yet do not match it. A Binary-ARGS variant with WVS-salience weighting (no Prospect Theory) is *worse* than uniform weighting in $12/15$ countries; the same variant with the Prospect-Theory utility recovers SWA-PTIS in $13/15$ countries—evidence that WVS weighting without gain/loss asymmetry is not enough. Full tables are in Appendices M–N. The EXP-24 headline models (Table 3) are evaluated under MIS on a disjoint 20-country list and should not be numerically compared cell-wise with Table 4.

### 5.3 Per-Country Grids and Failure Modes

The repository tracker (`exp_paper/tracker.md`) holds the full $20 \times 8$ MIS, JSD, MAE, flip-rate, and per-dimension worst-gap tables; reproducing them from `compare/comparison.csv` under `results/exp24_paper_20c/` is a single script away. Qualitatively, the pain points recur on **Species_Humans** and **Utilitarianism_More**: models still hug $\approx 50\%$ spare rates when humans are extreme, and dual-pass damping cannot invent information that the base logits never encode. Iran and Vietnam show occasional vanilla<DPBR inversions for otherwise strong checkpoints (Phi-4, Gemma-4-2B), which we attribute to IS noise and thin persona coverage rather than to a systematic geographic effect. Romania on Llama-3.2-1B is an outlier in the other direction (flip rate $>70\%$ in the logged run), highlighting that low parameter counts amplify variance in the reliability weight. Legacy 15-country JSD heatmaps for Qwen/Llama backbones remain in Appendices L–L for comparison with earlier SWA-PTIS drafts.

### 5.4 Robustness Suite at a Glance

Table 5 records the pre-DPBR robustness battery (JSD on the 15-country Llama/Qwen panel). The checks remain informative—temperature sweeps stay inside noise and the Prospect-Theory kernel dominates deterministic shifts—but they do not certify the newer 20-country MIS sweep. Treat them as supporting evidence for design choices carried forward into EXP-24; dual-pass reliability deserves its own seed and `VAR_SCALE` ablations (future work / supplement).

Table 5: Robustness suite. Each test independently probes one assumption behind SWA-DPBR; the bootstrap confidence interval on JSD is $\pm 0.004$.

| Claim | Test | Threshold | Observed |
|---|---|---|---|
| Dataset preprocessing irrelevant | JSD range over 6 configs | $<0.010$ | **0.0034** |
| $T_{\text{dec}}$ insensitivity | JSD span (Sweep A) | $<0.010$ | **0.0074** |
| Uniform $T_{\text{cat}}$ insensitivity | JSD span (Sweep B) | $<0.010$ | **0.0073** |
| Per-category $T_{\text{cat}}$ insensitivity | JSD span (Sweep C) | $<0.010$ | **0.0077** |
| PT-IS $>$ best deterministic shift | $\Delta$JSD(CS-Clamp$\rightarrow$SWA) | $>0$ | **0.003** |
| WVS weighting alone insufficient | ARGS-WVS $\succ$ ARGS-Unif | $<8/15$ | **3/15** |
| SWA-DPBR $\equiv$ ARGS-PT (kernel) | ARGS-PT best/tied | $\geq 10/15$ | **13/15** |
| Qwen2.5-32B failure: logit concentration | $H$ vs. $H_{\text{ref}}$=2.5 | $H<1$ | **0.521** nats |
| Cross-lingual pipeline generalises | end-to-end runs | 8/8 | **8/8** |

## 6 Ablation Study

We ablate four design choices on Qwen2.5-72B, USA (342 scenarios; Table 6).[1]

Row 1 shows that dropping PT-IS and keeping only persona consensus hurts JSD and Pearson $r$ materially. Row 2 matches the full system within numerical noise on this slice (Appendix I). Row 3 shows that **positional debiasing** still matters (+12.0% JSD in the archival log; Species preference inflates by 7.7pp without it). Row 4 removes the WVS-grounded persona panel (base prompt only): JSD rises sharply and Pearson $r$ collapses, confirming that survey personas are load-bearing, not decorative. The IS *flip rate* is only 6.0%, so the framework *modulates confidence* rather than reversing decisions (Appendix J).

## 7 Discussion

Taken together, the results tell a single story rather than a scatter of tables: SWA-DPBR is a lightweight *intervention* that turns culturally grounded persona variation into a bounded logit correction on every scenario, with bootstrap reliability shrinking the IS offset when the two passes disagree. On well-calibrated 70B-class models it achieves the largest, cleanest alignment gains without touching weights; when personas already agree, corrections tend to be small because consensus and IS draws align. At deployment the dominant cost is the batched multi-agent forward pass; the IS loop reuses cached scalars and adds negligible wall-clock time (Appendix J). The mechanism narrative is unusually sharp because the consensus-shift and Binary-ARGS families share our infrastructure but swap the decoding rule: uniform persona consensus (CS-MEAN, ARGS-UNIF) matches B5 PersonaConsensus exactly, so the lift is not "more personas in the prompt"; ARGS-WVS is worse than uniform in 12/15 countries, so WVS salience weighting without Prospect-Theory asymmetry is not the answer; CS-CLAMP recovers most of the gain, which pins bounded corrections as necessary; and the remaining $0.003$ JSD over CS-CLAMP is matched by ARGS-PT (JSD $0.0485$, $r$ $0.514$, best/tied in 13/15), i.e. by the same Prospect-Theory kernel we write into Eq. 7. Failure modes, by contrast, track *models*, not countries: Qwen2.5-32B degrades in 7/15 countries, lining up with the logit-concentration diagnosis in Section 5 and Appendix O.

**Limitations.** The caveats that most directly bound how far one should generalise from this paper are threefold. First, five hyperparameters ($\sigma$, $\lambda_{\text{coop}}$, $\eta$, $T_{\text{dec}}$, $T_{\text{cat}}$) were selected on 200 GPT-4-generated dilemmas with author proxy labels and transfer-checked on 50 MultiTP scenarios; the

[1]The ablation uses the full 342-scenario pool and yields JSD $= 0.0425$, vs. $0.0308$ in Table 10, because the per-country table uses a different scenario subset. The full system numbers are otherwise consistent.

Table 6: Ablation on Qwen2.5-72B, USA (342 scenarios), single-pass SWA-PTIS (pre-DPBR). Full system: JSD = 0.0425, $r$ = 0.724.

| # | Configuration | JSD ↓ | ΔJSD | $r$ ↑ | Δ$r$ | MAE |
|---|---|---|---|---|---|---|
| | **Full SWA-PTIS (archival)** | **0.0425** | – | **0.724** | – | **9.77** |
| 1 | No-IS (consensus only) | 0.0493 | +.007 | 0.650 | −.074 | 11.02 |
| 2 | Always-on PT-IS | 0.0422 | −.000 | 0.729 | +.005 | 9.74 |
| 3 | No debiasing | 0.0476 | +.005 | 0.743 | +.019 | 9.67 |
| 4 | Without persona | 0.0610 | +.019 | 0.441 | −.283 | 11.09 |

temperature and dataset sweeps in Appendices E–F are our main evidence that these values carry over. Second, we optimise against country-level AMCE vectors from Moral Machine / MultiTP crowdsourcing [Awad et al., 2018, Jin et al., 2025]; we do not run a human study asking whether SWA-DPBR outputs feel culturally appropriate in context [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not a proof of perceived legitimacy. Third, all headline metrics aggregate six moral dimensions; we do not yet report which dimensions improve or worsen. Heterogeneous $T_{\text{cat}}$ raises an open question about uneven correction across dimensions—systematic per-dimension analysis is our largest analytical gap. Additional limitations (single seed, noisy Pearson $r$, indirect WVS-to-trolley mapping, geographic and temporal coverage) are in Appendix Q.

**Broader impact and safeguards.** SWA-DPBR sits in a tension familiar from pluralistic alignment work [Sorensen et al., 2024]: it can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also bake in harmful majorities or survey stereotypes if deployed naively. We suggest four safeguards: governance so SWA-DPBR does not override safety-critical guardrails and domain experts review target cultures; periodic refresh of Wave 7-grounded personas as newer WVS data appear; audits of per-dimension AMCE shifts to catch entrenched bias; and optional clipping or penalty terms on sensitive dimensions inside $U_{\text{total}}$ to turn post-hoc review into decoding-time constraints. Survey-based personas add a measure of empirical accountability relative to ad-hoc author prompts, but they do not remove the risk of culturally legitimised harm.

## 8 Conclusion

Cross-cultural moral alignment is usually framed as a data and training problem; SWA-DPBR shows that a frozen model can still be steered toward country-level preferences when survey-grounded personas disagree, using a single PT-weighted importance-sampling step whose cooperative gain objective matches what behavioural decision theory already says about moral trade-offs [Kahneman and Tversky, 1979]. The headline is not a universal percentage gain across every model: the strongest evidence is on instruction-tuned 70B backbones, where there is room to move and logits are well-behaved; small quantised models improve directionally on smaller vanilla error; and Qwen2.5-32B illustrates that inference-time control has a *compatibility* requirement: *collapsed* decision logits break importance weighting no matter how carefully the utility is designed. Mechanistically, the ablations and stripped-down decoders leave little ambiguity: the Prospect-Theory gain/loss kernel on bounded perturbations is what remains when personas, WVS-only weighting, and deterministic shifts are held fixed. Looking ahead, adaptive $\sigma$ tied to per-model entropy, per-dimension outcomes, open-ended moral formats beyond binary logits, longitudinal WVS updates, and decoding-time ethical constraints are the natural next steps to turn this proof-of-concept into a deployable component of pluralistic alignment pipelines [Sorensen et al., 2024].

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# A AMCE Estimation Details

**Binary dimensions.** For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [Jin et al., 2025].

**Utilitarianism dimension.** Utilitarianism scenarios vary the *number* of lives on each side rather than character identity. Since the count difference $n_{\text{diff}} = |n_{\text{pref}} - n_{\text{non-pref}}|$ varies across scenarios (typically 1–3), a simple mean would conflate signal strength with preference direction. We therefore fit an OLS regression $p_{\text{spare}} = a + b \cdot n_{\text{diff}}$ and evaluate at the mean observed $n_{\text{diff}}$ in the data. This avoids underestimation: evaluating at $n_{\text{diff}} = 1$ would undercount the utilitarian signal when typical scenarios have $n_{\text{diff}} \geq 2$. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded as non-discriminative.

**Human AMCE scale.** The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

**Consistency.** Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

# B Persona Construction Details

## B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

## B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata as follows:

1. **Variable extraction:** For each respondent, we extract 8 value dimension scores from specific WVS items: gender egalitarianism (mean of Q58P, Q59P, Q60P), religious importance (Q6P), interpersonal trust (Q43P), moral permissiveness (mean of Q50, Q52P, Q54P), work centrality (Q5P), family importance (Q1P), personal autonomy (Q39P), meritocratic orientation (Q40P). All items use inverted scales where higher = more progressive/positive.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded.

3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Natural-language conversion:** Dimension means are converted to descriptors using fixed thresholds, listed in full in Table 7. Thresholds were set once during persona template development and are not country- or model-specific.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

Table 7: Complete WVS-dimension-to-descriptor threshold table used by the persona generator. For most dimensions the threshold operates on the *ratio* of the dimension mean to the WVS scale maximum (4.0); the exception is moral permissiveness, which operates on the raw 1–10 mean because its underlying items use a different scale. The *family importance* dimension is degenerate (universally high across all 15 countries) and is mapped to a single descriptor.

| Dimension (input) | Cut-off | Descriptor |
|---|---|---|
| Religiosity (ratio) | $> 0.85$ | deeply religious |
| | $> 0.70$ | moderately religious |
| | $> 0.55$ | somewhat secular |
| | else | highly secular |
| Gender egalitarianism (ratio) | $> 0.85$ | strongly gender-egalitarian |
| | $> 0.75$ | moderately gender-egalitarian |
| | $> 0.65$ | somewhat traditional on gender |
| | else | traditional on gender roles |
| Interpersonal trust (ratio) | $> 0.55$ | high interpersonal trust |
| | $> 0.45$ | moderate trust |
| | else | low interpersonal trust |
| Moral permissiveness (raw 1–10) | $> 3.5$ | morally permissive |
| | $> 3.0$ | moderately permissive |
| | $> 2.5$ | morally conservative |
| | else | morally strict |
| Personal autonomy (ratio) | $> 0.90$ | strongly values personal autonomy |
| | $> 0.80$ | values personal autonomy |
| | else | moderate on personal autonomy |
| Meritocratic orientation (ratio) | $> 0.95$ | strongly meritocratic |
| | $> 0.85$ | meritocratic |
| | else | egalitarian on income |
| Work centrality (ratio) | $> 0.90$ | work is central to identity |
| | $> 0.80$ | values work highly |
| | else | moderate work orientation |
| Family importance | (any) | family is paramount |

### B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

*Direct thematic overlap:* WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

*Indirect modulation:* Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*; e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-DPBR does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and the

IS stage selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

# C PT–IS, Sampling, and Dual-Pass Reliability

## C.1 Utility shape and the omitted explicit KL-on-$\epsilon$ term

The scalar decision gap $\delta$ acts as a one-dimensional state and the perturbations $\epsilon_k$ as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to $U_{\text{total}}$ in Eq. 7: `ImplicitSWAController` and `Exp24DualPassController` match Eq. 7 exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

## C.2 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Binary-ARGS and consensus-shift baselines in Appendix N provide complementary evidence that WVS-only linear rewards underperform the Prospect-Theory kernel; we omit utility-shape rows from Table 6 for clarity.

## C.3 Why split $K$ into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is intentionally smooth—unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-$K$ bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

## C.4 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. The two passes (original and swapped) are then averaged at the level of $p_{\text{spare}}$. This guarantees that any token-level bias that systematically favours one position cancels exactly, regardless of whether the bias originates from the lane label, the group label, or both.

# D Hyperparameter Validation

Non-PT hyperparameters ($\sigma$, $\lambda_{\text{coop}}$, $\eta$, $T_{\text{dec}}$, $T_{\text{cat}}$) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a

disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. $T_{\text{cat}}$ values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

## E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}[\text{Species}]=4.0$, $T_{\text{cat}}[\text{Gender}]=3.5$. All three sweeps yield JSD spans below $0.010$ ($0.0074,\ 0.0073,\ 0.0077$ respectively), and in every sweep the SWA-DPBR default (†) is strictly *more conservative* than the empirically best value ($\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 8: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

| Sweep | Setting | JSD ↓ | Pearson $r$ ↑ | ΔJSD |
|---|---|---|---|---|
| A: $T_{\text{dec}}$ | 0.10 | 0.0502 | 0.662 | +0.0013 |
| | 0.25 | 0.0491 | 0.658 | +0.0002 |
| | 0.50† | 0.0489 | 0.630 | - |
| | 0.75 | 0.0476 | 0.636 | −0.0013 |
| | 1.00 | 0.0466 | 0.641 | −0.0023 |
| | 2.00 (best) | 0.0428 | 0.627 | −0.0061 |
| | JSD span | | **0.0074** | |
| B: Uniform $T_{\text{cat}}$ | 0.5 | 0.0521 | 0.642 | +0.0052 |
| | 1.0 | 0.0492 | 0.673 | +0.0023 |
| | 1.5† | 0.0469 | 0.700 | - |
| | 2.0 | 0.0461 | 0.707 | −0.0008 |
| | 3.0 | 0.0454 | 0.707 | −0.0015 |
| | 4.0 | 0.0451 | 0.701 | −0.0018 |
| | 6.0 (best) | 0.0448 | 0.684 | −0.0021 |
| | JSD span | | **0.0073** | |
| C: Others $T_{\text{cat}}$ | 0.75 | 0.0526 | 0.566 | +0.0039 |
| | 1.00 | 0.0512 | 0.589 | +0.0025 |
| | 1.50† | 0.0487 | 0.634 | - |
| | 2.00 | 0.0475 | 0.660 | −0.0012 |
| | 3.00 | 0.0458 | 0.693 | −0.0029 |
| | 4.00 (best) | 0.0450 | 0.706 | −0.0037 |
| | JSD span | | **0.0077** | |

The monotonic JSD reduction with larger $T_{\text{dec}}$ reflects the role of $T_{\text{dec}}$ in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, $r$ peaks near $T_{\text{dec}}=1.0$ and declines at $2.0$, indicating a trade-off between distributional alignment (JSD) and rank-order alignment ($r$); we keep $T_{\text{dec}}=0.5$ as a deliberately conservative operating point.

## F Dataset Preprocessing and Sensitivity

**Pipeline.** Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category

capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

**Reproducible left/right assignment.** When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple $(sub_1, sub_2, g_1, g_2)$ modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 9: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and $< 0.010$ in 9/10 countries; the only exception is RUS at $0.013$, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

| Country | D0-Default | D1-NoAug | D2-Cap40 | D3-Cap120 | D4-NoFlip | D5-Strict |
|---|---|---|---|---|---|---|
| USA | 0.0655 | 0.0674 | 0.0625 | 0.0644 | 0.0643 | 0.0684 |
| DEU | 0.0465 | 0.0515 | 0.0492 | 0.0467 | 0.0444 | 0.0484 |
| CHN | 0.0393 | 0.0363 | 0.0385 | 0.0383 | 0.0409 | 0.0398 |
| JPN | 0.0304 | 0.0291 | 0.0319 | 0.0296 | 0.0273 | 0.0320 |
| BRA | 0.0492 | 0.0551 | 0.0479 | 0.0510 | 0.0520 | 0.0480 |
| VNM | 0.0592 | 0.0574 | 0.0631 | 0.0585 | 0.0578 | 0.0626 |
| GBR | 0.0614 | 0.0667 | 0.0608 | 0.0618 | 0.0597 | 0.0633 |
| KOR | 0.0371 | 0.0354 | 0.0378 | 0.0357 | 0.0349 | 0.0382 |
| RUS | 0.0383 | 0.0489 | 0.0389 | 0.0374 | 0.0383 | 0.0357 |
| NGA | 0.0522 | 0.0568 | 0.0506 | 0.0525 | 0.0514 | 0.0538 |
| **Mean** | **0.0479** | 0.0505 | 0.0481 | 0.0476 | 0.0471 | 0.0490 |
| $\Delta$ vs D0 | - | $+0.0025$ | $+0.0002$ | $-0.0003$ | $-0.0008$ | $+0.0011$ |

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the $\pm 0.004$ bootstrap CI, so reported results are not confounded by preprocessing choices.

## G Cross-Lingual Token Elicitation

**Native-language prompting system.** Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

**Token verification.** Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

# H Baseline Implementation Details

## H.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-DPBR.

**Sensitivity analysis and fairness considerations.** Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient $\alpha$, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search—varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement—could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-DPBR is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

## H.2 ARGS and Controlled Decoding: from Incompatible to Reformulated

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] were originally formulated with trained reward models, which in the cross-cultural setting would demand a separate reward model per country, unavailable for MultiTP. However, the binary forced-choice structure of MultiTP admits a natural reformulation in which the reward is a closed-form functional of the persona logit gaps $\{\delta_i\}$. We exploit this in Appendix N (**Binary-ARGS**) to construct three reward specifications (ARGS-UNIF, ARGS-WVS, ARGS-PT) that share SWA-DPBR's batched forward pass and differ only in the reward kernel. The strongest variant, ARGS-PT, recovers the SWA-DPBR kernel exactly, providing an independent route to the same optimum.

# I Detailed Ablation Analysis

**Without persona:** Row 4 removes all WVS-conditioned persona prompts (base-only moral elicitation on the same scenarios). The large JSD lift and Pearson $r$ drop in Table 6 match the expected loss of within-country diversity and numerosity anchoring from the persona ensemble.

**Debiasing:** Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

# J Trigger Mechanism and Latency

EXP-24 runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 6.0% in the archival USA ablation, indicating confidence modulation rather than wholesale decision reversal.

Mean latency: 730ms (SWA-DPBR) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; IS operates on cached scalars and contributes negligibly.

# K Germany Pearson $r$ Analysis

Germany's JSD remains unchanged (0.0603) while Pearson $r$ drops from 0.845 to 0.747. In 6D, Pearson $r$ is sensitive to rank perturbations: a single dimension swap reduces $r$ by $\sim$0.1 while JSD (depending on magnitudes, not ordering) stays constant. Rare IS corrections introduce minor rank perturbations without affecting distributional distance.

## L Per-Country Results for All Models

Tables 10–14 present the full per-country breakdown (vanilla and SWA-PTIS absolute JSD, Pearson $r$, and percentage improvement) for the five archival models. The 20-country MIS grids for EXP-24 live in `exp_paper/tracker.md`; the tables here pre-date that sweep and remain for JSD-level comparison.

Table 10: Per-country results for **Qwen2.5-72B-Instruct**.

| Country | Lang. | JSD ↓ | | Pearson $r$ ↑ | | JSD Improv. | |
|---|---|---|---|---|---|---|---|
| | | Van. | SWA | Van. | SWA | (Δ) | (%) |
| USA | en | .1066 | **.0308** | 0.400 | 0.822 | −.076 | +71.1 |
| Germany | de | .0603 | .0603 | 0.845 | 0.747 | +.000 | +0.0 |
| China | zh | .0513 | .0371 | 0.851 | 0.762 | −.014 | +27.8 |
| Japan | ja | .0986 | .0553 | 0.634 | 0.786 | −.043 | +43.9 |
| Brazil | pt | .0754 | .0703 | 0.349 | 0.529 | −.005 | +6.7 |
| Saudi Arabia | ar | .0778 | .0796 | 0.443 | 0.527 | +.002 | −2.3 |
| Vietnam | vi | .1302 | .0783 | 0.569 | 0.456 | −.052 | +39.9 |
| France | fr | .0789 | .0479 | 0.617 | 0.838 | −.031 | +39.4 |
| India | hi | .0600 | .0561 | 0.670 | 0.739 | −.004 | +6.5 |
| South Korea | ko | .0720 | .0513 | 0.355 | 0.645 | −.021 | +28.8 |
| Great Britain | en | .1057 | **.0294** | 0.419 | 0.800 | −.076 | +72.2 |
| Russia | ru | .0800 | .0629 | 0.623 | 0.784 | −.017 | +21.4 |
| Mexico | es | .0760 | .0534 | 0.605 | 0.545 | −.023 | +29.8 |
| Nigeria | en | .1103 | .0649 | 0.407 | 0.615 | −.046 | +41.2 |
| Australia | en | .1047 | **.0258** | 0.438 | 0.841 | −.079 | +75.4 |
| **Mean** | | **.0859** | **.0536** | 0.548 | 0.696 | −.032 | **+37.6** |

Table 11: Per-country results for **Llama-3.1-70B-Instruct**.

| Country | Lang. | JSD ↓ | | Pearson $r$ ↑ | | JSD Improv. | |
|---|---|---|---|---|---|---|---|
| | | Van. | SWA | Van. | SWA | (Δ) | (%) |
| USA | en | .1561 | .1234 | 0.012 | 0.197 | −.033 | +20.9 |
| Germany | de | .0709 | .0391 | −0.455 | 0.685 | −.032 | +44.9 |
| China | zh | .0459 | .0311 | 0.406 | 0.802 | −.015 | +32.3 |
| Japan | ja | .0457 | .0416 | 0.496 | 0.466 | −.004 | +9.0 |
| Brazil | pt | .0532 | .0656 | −0.370 | 0.348 | +.012 | −23.3 |
| Saudi Arabia | ar | .0539 | .0596 | 0.043 | −0.210 | +.006 | −10.7 |
| Vietnam | vi | .1546 | .0529 | −0.519 | −0.135 | −.102 | +65.8 |
| France | fr | .0500 | .0516 | 0.250 | 0.394 | +.002 | −3.0 |
| India | hi | .0442 | .0337 | −0.315 | 0.715 | −.010 | +23.6 |
| South Korea | ko | .0893 | .0436 | −0.272 | 0.551 | −.046 | +51.2 |
| Great Britain | en | .1568 | .0918 | 0.012 | 0.286 | −.065 | +41.5 |
| Russia | ru | .0449 | .0413 | 0.539 | 0.637 | −.004 | +7.9 |
| Mexico | es | .1265 | .0425 | −0.371 | 0.307 | −.084 | +66.4 |
| Nigeria | en | .1582 | .0988 | 0.109 | 0.497 | −.059 | +37.5 |
| Australia | en | .1551 | .1010 | 0.039 | 0.224 | −.054 | +34.9 |
| **Mean** | | **.0937** | **.0612** | −0.026 | 0.384 | −.033 | **+34.7** |

## M Consensus-Shift Baselines

To isolate the contribution of PT-IS importance-weighted sampling from the simpler benefit of using persona consensus at all, we implement four deterministic shift baselines applied directly to the scalar consensus gap $\bar{\delta}$, all sharing the same batched forward pass and two-pass debiasing as SWA-DPBR: CS-MEAN ($\delta_{\text{opt}} = \bar{\delta}$, equivalent to B5 PersonaConsensus), CS-STD ($\bar{\delta} \pm \sigma_{\text{agents}}$ in the consensus direction), CS-SCALED ($\bar{\delta}(1 + \alpha\,\text{Var})$), and CS-CLAMP ($\text{clip}(1.5\bar{\delta}, \pm 1)$).

CS-MEAN reproduces B5 PersonaConsensus exactly (cross-validating our ablation hierarchy). CS-STD *degrades* relative to CS-MEAN in 9/15 countries: variance amplification in the deterministic

Table 12: Per-country results for **Qwen2.5-7B-Instruct** (4-bit).

| Country | Lang. | JSD ↓ | | Pearson $r$ | | JSD Improv. | |
|---|---|---|---|---|---|---|---|
| | | Van. | SWA | Van. | SWA | ($\Delta$) | (%) |
| USA | en | .0531 | .0448 | 0.048 | 0.531 | −.008 | +15.5 |
| Germany | de | .0530 | .0522 | 0.207 | 0.212 | −.001 | +1.4 |
| China | zh | .0512 | .0455 | 0.401 | 0.397 | −.006 | +11.2 |
| Japan | ja | .0527 | .0540 | 0.490 | 0.449 | +.001 | −2.3 |
| Brazil | pt | .0445 | .0331 | 0.415 | 0.837 | −.011 | +25.8 |
| Saudi Arabia | ar | .0539 | .0443 | 0.234 | 0.265 | −.010 | +17.9 |
| Vietnam | vi | .0547 | .0472 | 0.259 | 0.263 | −.008 | +13.6 |
| France | fr | .0529 | .0442 | 0.259 | 0.315 | −.009 | +16.5 |
| India | hi | .0570 | .0524 | 0.466 | 0.183 | −.005 | +8.1 |
| South Korea | ko | .0492 | .0329 | 0.239 | 0.363 | −.016 | +33.2 |
| Great Britain | en | .0520 | .0302 | 0.140 | 0.483 | −.022 | +41.9 |
| Russia | ru | .0506 | .0452 | 0.373 | 0.491 | −.005 | +10.5 |
| Mexico | es | .0479 | .0384 | 0.038 | 0.360 | −.010 | +19.9 |
| Nigeria | en | .0551 | .0411 | 0.101 | 0.425 | −.014 | +25.3 |
| Australia | en | .0550 | .0431 | 0.170 | 0.452 | −.012 | +21.7 |
| **Mean** | | **.0522** | **.0426** | 0.252 | 0.402 | −.010 | **+18.4** |

Table 13: Per-country results for **Llama-3.1-8B-Instruct** (4-bit).

| Country | Lang. | JSD ↓ | | Pearson $r$ | | JSD Improv. | |
|---|---|---|---|---|---|---|---|
| | | Van. | SWA | Van. | SWA | ($\Delta$) | (%) |
| USA | en | .0470 | .0333 | −0.323 | 0.498 | −.014 | +29.1 |
| Germany | de | .0716 | .0529 | −0.101 | 0.382 | −.019 | +26.1 |
| China | zh | .0561 | .0457 | −0.082 | 0.388 | −.010 | +18.5 |
| Japan | ja | .0518 | .0519 | 0.545 | 0.395 | +.000 | −0.2 |
| Brazil | pt | .0732 | .0628 | −0.222 | 0.109 | −.010 | +14.2 |
| Saudi Arabia | ar | .0551 | .0513 | −0.252 | −0.230 | −.004 | +6.9 |
| Vietnam | vi | .0505 | .0423 | −0.330 | 0.303 | −.008 | +16.2 |
| France | fr | .0515 | .0612 | −0.610 | −0.220 | +.010 | −18.8 |
| India | hi | .0756 | .0564 | −0.471 | 0.347 | −.019 | +25.3 |
| South Korea | ko | .0522 | .0419 | −0.352 | −0.062 | −.010 | +19.7 |
| Great Britain | en | .0375 | .0255 | −0.065 | 0.457 | −.012 | +32.0 |
| Russia | ru | .0560 | .0434 | −0.230 | 0.593 | −.013 | +22.5 |
| Mexico | es | .0439 | .0383 | −0.079 | 0.344 | −.006 | +12.7 |
| Nigeria | en | .0474 | .0410 | −0.229 | 0.303 | −.006 | +13.5 |
| Australia | en | .0505 | .0363 | −0.156 | 0.328 | −.014 | +28.1 |
| **Mean** | | **.0533** | **.0436** | −0.247 | 0.271 | −.011 | **+22.6** |

shift actively hurts in high-disagreement regimes. CS-CLAMP is the strongest deterministic shift; its largest improvements occur in high inter-persona-variance countries (USA $-0.015$, AUS $-0.016$, GBR $-0.012$), exactly where Prospect Theory's asymmetric weighting is most active. The residual aggregate gap CS-CLAMP→SWA-DPBR ($\Delta$JSD $\approx 0.003$) is attributed to PT's gain/loss asymmetry, which no scalar shift can replicate.

## N Binary-ARGS Reward-Guided Decoding

We adapt ARGS [Khanov et al., 2024] to the binary forced-choice setting as *Binary-ARGS*: $K$ perturbations $\delta_k = \bar{\delta} + \epsilon_k$ ($\epsilon \sim \mathcal{N}(0, \sigma^2)$) are sampled and re-weighted by $\exp(r_k/\beta)$, where $r_k$ is one of three closed-form rewards. ARGS-UNIF uses a uniform directional reward ($\text{sign}(\bar{\delta}) \cdot \delta_k/N$), equivalent to importance-weighted consensus averaging; ARGS-WVS weights each persona by its WVS cultural salience; ARGS-PT replaces the linear reward by the Prospect-Theory utility identical to the SWA-DPBR kernel.

Three findings emerge. **(1)** ARGS-UNIF (JSD 0.0524) closely matches B5 PersonaConsensus, confirming uniform importance weighting reduces to consensus averaging. **(2)** ARGS-WVS (JSD 0.0544) is *worse* than ARGS-UNIF in 12/15 countries: WVS salience weighting without

Table 14: Per-country results for **Qwen2.5-32B-Instruct** (4-bit). SWA-DPBR degrades in 7/15 countries due to concentrated logit distributions.

| Country | Lang. | JSD ↓ | | Pearson $r$ | | JSD Improv. | |
|---|---|---|---|---|---|---|---|
| | | Van. | SWA | Van. | SWA | (Δ) | (%) |
| USA | en | .0447 | .0440 | 0.887 | 0.738 | −.001 | +1.5 |
| Germany | de | .0587 | .0534 | 0.893 | 0.803 | −.005 | +9.1 |
| China | zh | .0507 | .0574 | 0.827 | 0.818 | +.007 | −13.1 |
| Japan | ja | .0485 | .0592 | 0.846 | 0.805 | +.011 | −22.0 |
| Brazil | pt | .0717 | .0645 | 0.552 | 0.422 | −.007 | +9.7 |
| Saudi Arabia | ar | .0883 | .1015 | 0.704 | 0.625 | +.013 | −14.9 |
| Vietnam | vi | .0900 | .0780 | 0.634 | 0.556 | −.012 | +13.1 |
| France | fr | .0730 | .0681 | 0.647 | 0.652 | −.005 | +6.7 |
| India | hi | .0737 | .0840 | 0.458 | 0.583 | +.010 | −13.8 |
| South Korea | ko | .0529 | .0303 | 0.106 | 0.064 | −.023 | +43.0 |
| Great Britain | en | .0470 | .0345 | 0.889 | 0.732 | −.013 | +26.5 |
| Russia | ru | .0700 | .0780 | 0.721 | 0.711 | +.008 | −11.3 |
| Mexico | es | .0775 | .0933 | 0.632 | 0.381 | +.016 | −20.3 |
| Nigeria | en | .0641 | .0512 | 0.124 | 0.156 | −.013 | +20.1 |
| Australia | en | .0541 | .0425 | 0.883 | 0.781 | −.012 | +21.4 |
| **Mean** | | **.0632** | **.0633** | 0.625 | 0.550 | +.000 | **−0.2** |

Table 15: Per-country JSD for the four consensus-shift baselines on Llama-3.1-70B (15 countries). CS-CLAMP is the strongest deterministic shift but still loses to SWA-DPBR in aggregate; CS-STD is catastrophic for NGA (+0.017 vs. CS-MEAN) due to unbounded variance amplification.

| Country | CS-Mean | CS-Std | CS-Scaled | CS-Clamp |
|---|---|---|---|---|
| USA | 0.0923 | 0.0919 | 0.0933 | **0.0771** |
| DEU | **0.0410** | 0.0448 | 0.0449 | 0.0428 |
| CHN | 0.0591 | 0.0618 | 0.0608 | **0.0520** |
| JPN | **0.0306** | 0.0335 | 0.0343 | 0.0311 |
| BRA | 0.0524 | 0.0519 | 0.0530 | **0.0459** |
| SAU | 0.0565 | 0.0584 | 0.0570 | **0.0536** |
| VNM | 0.0632 | 0.0659 | 0.0669 | **0.0564** |
| FRA | 0.0547 | 0.0516 | 0.0552 | **0.0442** |
| IND | 0.0421 | **0.0410** | 0.0413 | 0.0418 |
| KOR | 0.0451 | 0.0454 | 0.0451 | **0.0443** |
| GBR | 0.0800 | 0.0842 | 0.0844 | **0.0681** |
| RUS | 0.0450 | **0.0435** | 0.0443 | 0.0429 |
| MEX | 0.0539 | 0.0539 | 0.0531 | **0.0472** |
| NGA | 0.0826 | 0.0998 | 0.0987 | **0.0799** |
| AUS | 0.0868 | 0.0864 | 0.0861 | **0.0711** |
| **Mean** | 0.0590 | 0.0609 | 0.0612 | **0.0532** |
| **SWA-DPBR** | **0.0502** (Pearson $r$ = 0.424) | | | |

asymmetric loss-sensitivity actively misaligns persona contributions. **(3)** ARGS-PT (JSD $0.0485$, $r$ $0.514$) is best/tied in 13/15 countries and dominates on every aggregate metric, empirically showing that SWA-DPBR is precisely an ARGS variant with the same cooperative PT kernel as Eq. 7. The two countries where ARGS-PT underperforms (BRA, IND) have the highest inter-persona variance, where importance weights become noisy and corrections can overshoot.

## O Entropy-Aware Adaptive $\sigma$ for Qwen2.5-32B

We study whether an entropy-aware adaptive $\sigma$ can address the Qwen2.5-32B failure mode (Section 5). At each scenario we measure the Shannon entropy $H$ of the top-50 token distribution at the decision position and set $\sigma = \sigma_{\text{base}} \cdot \text{clip}(H/H_{\text{ref}},\ 0.2,\ 5.0)$ with $H_{\text{ref}}$=2.5 nats and $\sigma_{\text{base}}$=0.30. We compare against a fixed-$\sigma$ sweep on 8 countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR).

The mean entropy is $H = 0.521$ nats, far below $H_{\text{ref}}$, confirming concentrated logits. Adaptive $\sigma$ self-scales to $\bar{\sigma} = 0.086$ ($r(H, \sigma)$=0.948), validating the diagnostic but collapsing onto the low-$\sigma$ regime. The best fixed $\sigma$=0.80 reaches JSD $0.0599$ vs. $0.0613$ at default—a marginal absolute gain—while the per-country breakdown exposes the deeper problem: countries cluster into a high-$\sigma$

Table 16: Binary-ARGS per-country results (Llama-3.1-70B, 15 countries). ARGS-PT achieves best or tied JSD in 13/15 countries and best $r$ in 11/15, confirming SWA-DPBR is precisely an ARGS variant with the same distance-based Prospect-Theory utility as Eq. 7.

| | JSD ↓ | | | Pearson $r$ ↑ | | |
|---|---|---|---|---|---|---|
| Country | Unif | WVS | PT | Unif | WVS | PT |
| USA | 0.0757 | 0.0793 | **0.0646** | 0.506 | 0.484 | **0.626** |
| DEU | 0.0476 | 0.0512 | **0.0465** | 0.668 | 0.656 | **0.732** |
| CHN | 0.0347 | 0.0396 | 0.0362 | 0.753 | 0.721 | **0.788** |
| JPN | 0.0308 | 0.0311 | **0.0275** | 0.782 | 0.757 | **0.831** |
| BRA | **0.0482** | 0.0509 | 0.0519 | 0.351 | 0.317 | **0.375** |
| SAU | 0.0541 | 0.0539 | 0.0534 | 0.076 | 0.079 | **0.165** |
| VNM | 0.0717 | 0.0755 | **0.0551** | −0.507 | −0.427 | −0.369 |
| GBR | 0.0692 | 0.0709 | **0.0588** | 0.494 | 0.488 | **0.616** |
| KOR | 0.0421 | 0.0386 | 0.0364 | 0.722 | **0.750** | 0.748 |
| RUS | **0.0399** | 0.0443 | 0.0415 | 0.723 | 0.728 | **0.782** |
| MEX | 0.0464 | 0.0496 | **0.0428** | 0.255 | 0.256 | **0.503** |
| NGA | 0.0567 | 0.0598 | **0.0502** | 0.775 | 0.779 | **0.852** |
| AUS | 0.0742 | 0.0763 | **0.0664** | 0.461 | 0.452 | **0.557** |
| FRA | 0.0495 | 0.0484 | 0.0480 | 0.336 | 0.370 | **0.423** |
| IND | **0.0457** | 0.0463 | 0.0475 | 0.045 | −0.003 | **0.079** |
| **Mean** | 0.0524 | 0.0544 | **0.0485** | 0.429 | 0.427 | **0.514** |

Table 17: Per-country JSD under fixed-$\sigma$ on Qwen2.5-32B. Two latent country clusters emerge: USA/JPN/GBR/KOR strongly prefer high $\sigma$ (USA: $0.0523 \rightarrow 0.0317$ at $\sigma$=1.0); DEU/CHN/BRA/VNM strictly prefer low $\sigma$. All 8 countries share $H \approx 0.52$ nats, so the entropy signal alone cannot discriminate them.

| $\sigma$ | USA | DEU | CHN | JPN | BRA | VNM | GBR | KOR | Mean |
|---|---|---|---|---|---|---|---|---|---|
| 0.05 | 0.0515 | 0.0647 | 0.0533 | 0.0690 | 0.0692 | 0.0710 | 0.0500 | 0.0670 | 0.0620 |
| 0.10 | 0.0519 | 0.0648 | 0.0532 | 0.0684 | 0.0694 | 0.0711 | 0.0499 | 0.0668 | 0.0619 |
| 0.15 | 0.0522 | 0.0649 | 0.0530 | 0.0676 | 0.0698 | 0.0711 | 0.0499 | 0.0665 | 0.0619 |
| 0.20 | 0.0518 | 0.0651 | 0.0527 | 0.0661 | 0.0702 | 0.0713 | 0.0498 | 0.0661 | 0.0616 |
| $0.30^{\dagger}$ | 0.0523 | 0.0654 | 0.0516 | 0.0644 | 0.0710 | 0.0717 | 0.0491 | 0.0649 | 0.0613 |
| 0.50 | 0.0487 | 0.0661 | 0.0515 | 0.0601 | 0.0732 | 0.0734 | 0.0476 | 0.0604 | 0.0601 |
| 0.80 | **0.0427** | 0.0672 | 0.0599 | **0.0585** | 0.0780 | 0.0757 | **0.0439** | **0.0532** | **0.0599** |
| 1.00 | 0.0317 | 0.0681 | 0.0714 | 0.0637 | 0.0833 | 0.0744 | 0.0458 | 0.0522 | 0.0613 |
| Adaptive ($\bar{\sigma}$=0.086) | 0.0516 | 0.0647 | 0.0532 | 0.0688 | 0.0693 | 0.0711 | 0.0500 | 0.0670 | 0.0620 |

group (USA/JPN/GBR/KOR) and a low-$\sigma$ group (DEU/CHN/BRA/VNM), but *all* countries share $H \approx 0.52$ nats, so a per-scenario entropy signal cannot discriminate them. A complete fix therefore requires richer per-country proposal adaptation—left as future work.

# P Extended Country Pipeline Run (Engineering Check)

We run the SWA-DPBR pipeline end-to-end on 8 additional countries not in the MultiTP benchmark, spanning 4 script systems and 5 language families: Poland (pl), Turkey (tr), Indonesia (id), Argentina (es), Ukraine (uk), Taiwan (zh), the Philippines (tl), and Egypt (ar). **This appendix makes an engineering claim, not a scientific one.** Because no Moral Machine human AMCEs are available for these countries, we evaluate on 300 synthetic scenarios per country whose Vanilla AMCE is uniformly $50.0$ by construction. The only thing this experiment establishes is that the persona-construction, translation, batched-forward-pass, and IS-update code runs to completion across the 4 scripts and 5 language families above. The $\sim$+0.05pp "AMCE shift" is therefore measuring perturbation activity, not alignment, and the mean JSD numbers should not be compared to the main table. We retain the appendix only as evidence that the framework is not silently English-specific.

Because the synthetic scenarios uniformly initialise Vanilla AMCE at $50.0$, the shift metric measures only the pipeline's ability to modulate output distributions away from an uninformative prior; it cannot be interpreted as alignment improvement over human preferences and is not directly comparable to the main 15-country results. We include this probe purely to demonstrate that the WVS-grounded persona construction, the batched forward pass, the two-pass debiasing, and the IS loop all run end-to-end across right-to-left Arabic, CJK, Cyrillic, and several Latin-script languages without any

Table 18: Extended country pipeline run (Llama-3.1-70B). The pipeline executes end-to-end for 4 script systems and 5 language families; IS-induced AMCE shifts are detected in all 8 countries (mean $+0.05 \pm 0.02$ pp).

| Country | Lang. | Script | AMCE Shift | Vanilla JSD | SWA JSD |
|---|---|---|---|---|---|
| Poland (POL) | Polish | Latin | +0.03 pp | 0.0496 | 0.0497 |
| Turkey (TUR) | Turkish | Latin | +0.04 pp | 0.0376 | 0.0377 |
| Indonesia (IDN) | Indonesian | Latin | +0.06 pp | 0.0425 | 0.0424 |
| Argentina (ARG) | Spanish | Latin | +0.04 pp | 0.0388 | 0.0387 |
| Ukraine (UKR) | Ukrainian | Cyrillic | +0.07 pp | 0.0485 | 0.0483 |
| Taiwan (TWN) | Chinese | CJK | +0.03 pp | 0.0427 | 0.0426 |
| Philippines (PHL) | Filipino | Latin | +0.06 pp | 0.0396 | 0.0396 |
| Egypt (EGY) | Arabic | Arabic | +0.06 pp | 0.0511 | 0.0513 |
| **Mean** | | | $\mathbf{+0.05 \pm 0.02}$ **pp** | 0.0438 | 0.0438 |

code change. Future work should collect human AMCE annotations for these countries to enable full quantitative evaluation.

## Q Extended Limitations

**Single seed and stochasticity.** All reported archival numbers use one run per model with one IS random draw ($K$=128 perturbations) per scenario; variance across seeds is not characterised here.

**Pearson $r$ in six dimensions.** As noted in Section 5, the AMCE vector is short, so per-country $r$ has large sampling error and small row-wise $r$ deltas are not necessarily statistically separable; JSD remains the primary claim-bearing metric.

**WVS-to-trolley linkage.** The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension $X$ caused AMCE shift $Y$."

**Inter-persona reward assumption.** The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed.

**Geographic coverage, preprocessing, and survey vintage.** We evaluate 15 of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

**Binary format.** The logit-difference formulation requires generalisation to softmax for multi-option settings.

**Language–culture conflation.** One language per country may conflate linguistic and cultural effects.

**Model quantisation.** 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

**WVS temporal staleness.** Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

**Qwen2.5-32B failure mode.** Concentrated logit distributions ($H$=0.521 nats) cause IS perturbations to land off-manifold and yield near-uniform importance weights. Appendix O shows that an entropy-aware $\sigma$ correctly diagnoses the cause but cannot fix it on its own, because all 8 evaluated

countries share the same low entropy and thus cannot be discriminated by a per-scenario entropy signal. A complete remedy requires richer jointly adaptive $\sigma$ (and related proposal knobs) per country and is left to future work.